strongly_connected_component_subgraphs

`strongly_connected_component_subgraphs`(*G*, *copy=True*) [source]

Generate strongly connected components as subgraphs.

Parameters:
- **G** (*NetworkX Graph*) – A directed graph.
- **copy** (*boolean, optional*) – if copy is True, Graph, node, and edge attributes are copied to the subgraphs.

Returns: **comp** – A generator of graphs, one for each strongly connected component of G.

Return type: generator of graphs

Examples

Generate a sorted list of strongly connected components, largest first.

```
>>> G = nx.cycle_graph(4, create_using=nx.DiGraph())
>>> G.add_cycle([10, 11, 12])
>>> [len(Gc) for Gc in sorted(nx.strongly_connected_component_subgraphs(G),
...                         key=len, reverse=True)]
[4, 3]
```

If you only want the largest component, it's more efficient to use max instead of sort.

```
>>> Gc = max(nx.strongly_connected_component_subgraphs(G), key=len)
```

See also

`connected_component_subgraphs()`, `weakly_connected_component_subgraphs()`